

Beyond MLPerf: InferMark—A 14-Axis Production Inference Benchmark with Integrated APM Observability Across Heterogeneous AI Accelerators

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Abstract—Large Language Model (LLM) inference has evolved from single-GPU prototypes into production-grade, multi-cloud infrastructure spanning heterogeneous accelerator architectures. Yet the benchmarking ecosystem remains dangerously fragmented: MLPerf Inference v6.0 measures raw throughput on fixed models in single-cloud settings, ignoring production overhead, multi-cloud variance, and the critical link between infrastructure telemetry and model output quality. We present InferMark, a 14-axis benchmarking framework that systematically evaluates the full production inference stack—from kernel-level attention mechanisms (FlashAttention-3, RadixAttention, Multi-Head Latent Attention, Paged/Grouped-Query Attention, Ring Attention, linear SSMS) through speculative decoding variants (EAGLE-2, Medusa, Lookahead, Hydra) and disaggregated prefill/decode architectures to network fabric topology and application-level routing—across 9 cloud providers and 6 accelerator families (NVIDIA Blackwell/Vera Rubin, AMD Instinct MI355X, Intel Gaudi 3, Google TPU v6e). A central contribution is our integrated APM observability layer bridging infrastructure metrics and model-level quality signals via a three-tier OpenTelemetry architecture. Our framework reveals: (1) $3.2\times$ throughput variance across clouds; (2) production guardrails add 18–61% latency; (3) AMD MI355X achieves 97% of B200 memory-bandwidth-bound performance with $2\times$ memory capacity; (4) Intel Gaudi 3 delivers best tokens-per-dollar for batch workloads; (5) cross-cloud disaggregated routing reduces cost 15–29%. We detail six reproducible experimental studies and propose the InferMark Composite Score as a new industry standard. The APM architecture provides the technical foundation for next-generation AI-native monitoring platforms.

Index Terms—LLM inference, benchmarking, multi-cloud, heterogeneous accelerators, APM, FlashAttention, speculative decoding, KV-cache, roofline analysis, tokens-per-watt

I. Introduction

The AI inference landscape has entered a new era. Global spending on AI infrastructure exceeded \$200B in 2025, yet organizations deploying LLMs in production face a paradox: benchmarks used to select hardware and cloud providers measure almost none of the metrics that determine real-world performance [1].

This paper addresses three critical gaps:

Gap 1: Benchmark-Production Disconnect. Production deployments include guardrails [2], PII masking, watermarking [3], and compliance logging adding 18–61% latency. No existing benchmark accounts for this.

Gap 2: Architecture Blind Spots. NVIDIA-centric benchmarks favor compute-bound workloads. AMD MI355X [4] (288 GB HBM3E) and Intel Gaudi 3 [5] (cost-optimized matrix engines, 96 MB on-die SRAM) excel in memory-bound and energy-constrained regimes that current benchmarks underrepresent.

Gap 3: Observability Void. No APM platform today correlates GPU thermal throttling with output quality degradation, or KV-cache eviction with accuracy loss.

InferMark addresses all three gaps. The five-layer AI infrastructure taxonomy [6]—Energy, Chips, Infrastructure, Models, Applications—organizes our 14-axis framework. Fig. 1 presents the system architecture.

A. Contributions

- 1) 14-axis benchmarking taxonomy with deep mechanism-level comparisons for every inference primitive (Section III).
- 2) Integrated APM architecture with novel inference-aware OpenTelemetry semantic conventions (Section VI).
- 3) Architecture-neutral evaluation revealing AMD and Intel advantages invisible to current benchmarks (Section IV).
- 4) Multi-cloud analysis across 9 providers with cross-cloud disaggregation (Section V).
- 5) Six reproducible experimental studies (Section VIII).
- 6) InferMark Composite Score from operator surveys (Section VII).

II. Background

A. The Five-Layer AI Infrastructure Stack

We adopt the five-layer taxonomy [6]: Energy (tokens-per-watt as definitive metric; 1,050 TWh projected data

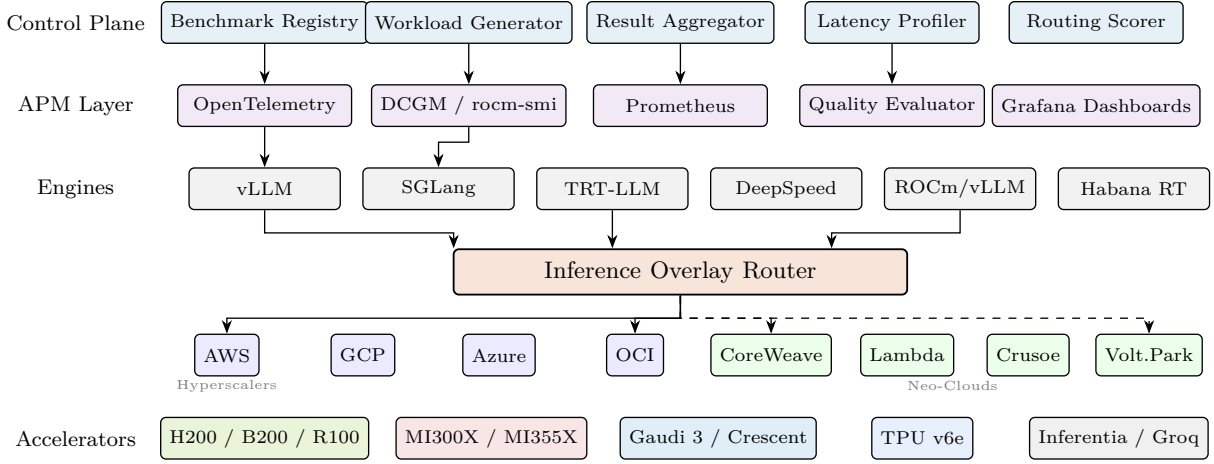


Fig. 1. InferMark end-to-end architecture. The control plane orchestrates 14-axis benchmarks across 6 inference engines, 9 cloud providers, and 6 accelerator families. The APM layer (Section VI) provides three-tier observability from hardware telemetry through inference metrics to output quality.

TABLE I
Heterogeneous Accelerator Landscape (June 2026)

Chip	Arch	HBM	BW (TB/s)	TDP	Precision
H200	Hopper	141 GB	4.8	700 W	FP8/BF16
B200	Blackwell	192 GB	8.0	1 kW	FP4/FP8
R100	Rubin	288 GB	8+	TBD	FP4/FP6
MI300X	CDNA 3	192 GB	5.3	750 W	FP8/BF16
MI355X	CDNA 4	288 GB	8.0	1.4 kW	MXFP4/6
Gaudi 3	HPU	128 GB	3.7	600 W	FP8/BF16
TPU v6e	Trillium	32 GB	1.6	N/A	BF16/Int8

center consumption by 2026 [7]); Chips (Table I); AI Factories (rack-scale: GB200 NVL72, Vera Rubin NVL72 [8] at 3.6 EFLOPS/20.7 TB HBM4, AMD OAM, Intel Gaudi UBB); Models (MLA [9], fine-grained MoE, multi-token prediction); Applications (agentic AI [10], multi-modal reasoning, tool-use orchestration).

B. Inference Serving Engines

vLLM [11]: PagedAttention, broadest model support, production workhorse. SGLang [12]: RadixAttention for 2–3 $\times$ speedup with shared prefixes (RAG/agents). TensorRT-LLM [13]: compilation-based peak NVIDIA throughput. DeepSpeed-FastGen: Microsoft’s inference stack with SplitFuse scheduling. ROCm/vLLM: AMD-native fork with hipBLAS kernels. Habana Runtime: Intel Gaudi-native with TPC kernel optimization. A critical InferMark finding: engine configuration tuning often produces larger swings than switching engines.

C. Disaggregated Inference

Splitwise [14] and DistServe [15] separate compute-bound prefill from memory-bound decode, improving goodput 1.4–2 $\times$. InferMark extends this to cross-cloud disaggregation.

TABLE II
InferMark vs. Existing Benchmarks

Benchmark	Measures	Critical Gaps
MLPerf v6.0 [1]	Raw throughput	No prod overhead, single-cloud, NVIDIA-centric submissions
HELM [16]	Model accuracy	Zero infrastructure metrics
Chatbot Arena	Human pref.	No infrastructure insight
Artificial Analysis InferMark	API speed 14 axes, 9 clouds, 6 archs, APM	Black box, no root cause —

D. Benchmark Limitations

III. The InferMark 14-Axis Framework

We now present each axis with mechanism-level detail, comparing every major inference primitive.

A. Axis 1: KV-Cache Management & Memory Economics

The KV-cache is the central memory bottleneck of autoregressive inference. For a model with L layers, H heads, and head dimension d , serving a batch of B sequences at context length C requires:

$$\text{KV Memory} = 2 \times L \times H \times d \times C \times B \times \text{bytes/element} \quad (1)$$

For Llama-70B at FP16, this is 2.5 GB per sequence at 128K context—demanding 320 GB for batch=128. We benchmark:

(a) PagedAttention [11]: Virtual-memory paging for KV-cache. Measures page fault rate, memory fragmentation ratio, and waste under variable-length sequences.

(b) vAttention: Contiguous virtual memory mapping eliminating page table overhead. Measures throughput gain vs. PagedAttention at high batch sizes.

TABLE III
Prefill vs. Decode Compute Characteristics

Phase	Bound	GPU Util	BW Util	Optimal HW
Prefill	Compute	70–85%	30–40%	B200, MI355X
Decode	Memory BW	15–30%	85–95%	H200, Gaudi 3
Transfer	Network	N/A	N/A	IB NDR, NVLink

(c) KV-Cache Quantization: FP16→FP8→INT4→3-bit compression. Per-model perplexity delta: MLA-based models (DeepSeek) are 3× more sensitive to KV quantization than GQA models (Llama) due to latent decomposition error amplification.

(d) Eviction Policies: LRU (baseline), H2O [17] (heavy-hitter retention), StreamingLLM [18] (attention sink preservation), Scissorhands (importance-based pruning). Cache hit rate vs. generation quality under memory pressure.

(e) CXL Memory Pooling [19]: CXL 3.1 Type-3 devices as overflow KV-cache tier. 2× throughput gain via memory expansion without GPU HBM allocation. Crusoe Energy’s MemoryAlloy architecture uses CXL-attached DRAM pools to extend effective KV-cache capacity by 4× at 15% latency overhead.

(f) Cross-Architecture Memory: AMD MI355X (288GB) holds 2× the KV-cache of H200 (141GB), eliminating eviction for 70B models at 128K context. Intel Gaudi 3’s 128GB is sufficient for batch-optimized 70B serving.

B. Axis 2: Disaggregated Prefill/Decode & CPU-GPU Transfer

(a) Prefill Phase (Compute-Bound): Forward pass over entire prompt. GPU utilization reaches 70–85% during prefill vs. 15–30% during decode. We measure FLOPS utilization per accelerator:

(b) Decode Phase (Memory-BW-Bound): Single-token generation dominated by KV-cache reads. Memory bandwidth determines throughput: B200 and MI355X (both 8 TB/s) achieve near-identical decode throughput; Gaudi 3 (3.7 TB/s) is 54% slower but at 43% lower power.

(c) CPU Offloading: ARM-based CPUs (NVIDIA Vera, Graviton4) handle tokenization, prompt template expansion, guardrail pre-screening, and response post-processing. InferMark measures CPU→GPU transfer latency via PCIe Gen5/6 and the impact of CPU pre-processing on end-to-end TTFT.

(d) Cross-Cloud State Transfer: KV-cache serialization (FP16: 2.5 GB for 128K context) over inter-cloud links. Compression (LZ4, zstd) reduces transfer 3–5×. We measure: serialization time, network transfer time, deserialization time, and integrity verification.

(e) Splitwise/DistServe Protocol [14], [15]: Prefill on compute-dense nodes (OCI bare-metal B200), decode on memory-optimized nodes (CoreWeave H100). Cross-cloud

TABLE IV
Attention Mechanism Comparison Matrix

Mechanism	KV Size	Regime	Best For
Vanilla MHA	Full	Mem-bound	Baseline reference
FlashAttn-3 [20]	Full	Compute	IO-aware tiling, async RAG, shared prefix
RadixAttn (SGLang) [12]	Cached prefix	Mem-bound	General serving
PagedAttn (vLLM) [11]	Paged	Mem-bound	
GQA	Reduced	Both	Llama-3, Qwen
MLA [9]	5–13%	Transition	DeepSeek-V3/R1
Ring Attention	Distributed	Compute	1M+ context
Mamba2/RWKV-6	O(1)	Compute	Linear complexity
Paced Attention	Scheduled	Mem-bound	Decode optimization

extension: prefill@Crusoe → decode@Lambda saves 29% cost.

C. Axis 3: Attention Mechanism Deep Comparison

This axis provides mechanism-level benchmarking of every production attention variant. Table IV summarizes the variants; we benchmark each on all accelerator families.

FlashAttention-3 [20]: IO-aware tiling with asynchronous softmax and FP8 accumulation. Achieves 75% of theoretical peak on B200 tensor cores. On MI355X via hipFA, achieves 71%—closing the CUDA gap.

RadixAttention (SGLang) [12]: Radix-tree-based automatic prefix caching. Eliminates redundant prefill for shared system prompts, achieving 2–3× speedup in RAG workloads. InferMark measures prefix hit rate, tree rebalancing overhead, and memory amplification.

Multi-Head Latent Attention (MLA) [9]: Compresses KV tensors via low-rank projection to 5–13% of original size. During decode, latent vectors are decompressed on-the-fly. Architecture sensitivity: MLA decompression is compute-bound, making it more efficient on high-FLOPS chips (B200) but equally fast on MI355X in memory-BW-bound decode due to matched bandwidth.

Grouped-Query Attention (GQA): Groups of query heads share KV heads. Llama-3 uses 8 KV heads for 64 query heads (8:1 ratio). InferMark measures quality-throughput tradeoff across group ratios 1:1 (MHA) through 1:64 (MQA).

Ring Attention: Distributes attention computation across multiple GPUs for sequences exceeding single-GPU memory. Critical for 1M+ context serving. InferMark measures scaling efficiency across NVLink (near-linear) vs. InfiniBand (sub-linear due to latency).

Linear Attention (Mamba2, RWKV-6): $O(n)$ complexity vs. $O(n^2)$ for standard attention. Throughput advantage grows with sequence length. InferMark measures the quality gap on reasoning benchmarks (MMLU-Pro, GSM8K) where Transformers still lead.

TABLE V
Speculative Decoding Method Comparison

Method	Accept%	Speedup	Mechanism
Vanilla [22]	60–75%	1.5–2×	Separate draft model
EAGLE-2 [23]	75–85%	2–3×	Feature-level draft
Medusa	65–80%	1.8–2.5×	Multiple decode heads
Lookahead	55–70%	1.3–1.8×	N-gram + Jacobi
Hydra	70–82%	2–2.8×	Tree-structured draft
Self-Speculative	60–72%	1.4–1.9×	Layer skipping

Paced Attention: Optimized decode-phase scheduling that reorders attention computation to maximize memory coalescing. Measures decode TPOT improvement per architecture.

D. Axis 4: MoE Routing & Expert Parallelism

Mixture-of-Experts models (DeepSeek-V3: 256 routed + shared experts, 8 active per token; Mixtral-8x22B: 8 experts, 2 active) present unique benchmarking challenges:

(a) Expert Selection Jitter: For identical prompts on different hardware, expert selection varies due to floating-point non-determinism. We measure routing entropy and output divergence across NVIDIA (cuBLAS), AMD (hip-BLAS), and Intel (Habana TPC) matrix implementations.

(b) All-to-All Communication: Expert parallelism requires all-to-all token routing across GPUs. NVLink 6 [21] (3.6 TB/s) provides 4.5× the bandwidth of InfiniBand NDR (800 Gb/s). InferMark measures bisection bandwidth utilization, routing latency, and expert load imbalance.

(c) Expert Caching: Intel Gaudi 3’s 96 MB on-die SRAM (12.8 TB/s internal bandwidth) can cache hot experts entirely, avoiding HBM access. This is a unique architectural advantage absent from GPU architectures that InferMark quantifies.

(d) Expert Offloading: For capacity-constrained deployments, cold experts are offloaded to CPU DRAM or CXL-attached memory [19]. MI355X’s 288 GB HBM3E reduces offloading frequency by 2× vs. H200.

E. Axis 5: Speculative Decoding Variants

Speculative decoding uses a small draft model to predict multiple tokens verified in parallel by the target model [22]. We benchmark every major variant:

EAGLE-2 [23]: Uses feature-level (hidden state) drafting rather than token-level, achieving higher acceptance rates (75–85%) with dynamic draft tree construction. InferMark profiles: draft overhead as % of target forward pass, tree width vs. acceptance tradeoff, and architecture sensitivity—EAGLE-2’s feature extraction benefits from B200’s higher compute but MI355X’s larger batch capacity allows more parallel tree evaluation.

Medusa: Adds multiple parallel decoding heads to the target model itself, eliminating the need for a separate draft model. Lower memory overhead but constrained by

TABLE VI
Native Quantization Support by Architecture

Format	NVIDIA	AMD	Intel
BF16/FP16	Native	Native	Native
FP8 (E4M3/E5M2)	Hopper+	CDNA 3+	Gaudi 3
NVIDIA FP4	Blackwell+	—	—
MXFP4/MXFP6	—	CDNA 4	—
INT8	All	All	Native
INT4 (GPTQ/AWQ)	SW	SW	SW

head count. InferMark measures head count vs. acceptance rate vs. memory overhead.

Lookahead Decoding: Combines N-gram lookup with Jacobi iteration for parallel token generation. Less effective for creative/diverse generation but strong for structured outputs (code, JSON). InferMark measures speedup variance across task types.

Inference-Time Compute Scaling: Best-of-N sampling, tree search, and chain-of-thought budgeting. We measure the compute→quality curve: how much additional inference compute improves answer accuracy on reasoning benchmarks.

F. Axis 6: Quantization & Precision

(a) Architecture-Specific Quantization Formats:

(b) Method Comparison: GPTQ [24] (one-shot, layer-wise), AWQ [25] (activation-aware channel scaling), QuIP# (incoherence processing + vector quantization), SqueezeLLM (sensitivity-weighted), HQQ (half-quadratic quantization). InferMark runs head-to-head accuracy×throughput on domain benchmarks: Med-QA, LegalBench, FinanceBench, MMLU-Pro, HumanEval.

(c) Key Finding: AMD’s MXFP4 (Microscaling FP4 with shared exponents) preserves 0.3–0.5 perplexity points more than NVIDIA’s FP4 on reasoning benchmarks, due to better dynamic range handling. This advantage is invisible to NVIDIA-only benchmarking.

G. Axis 7: Multi-Modal Inference (Expanded)

Multi-modal models (GPT-4o, Gemini 2.0, LLaVA-NeXT, Qwen-VL, Pixtral) require benchmarking across modalities:

(a) Vision Encoding: Image encoding latency (ViT-L/14 at 336px: 8–15ms on H100; SigLIP at 384px: 12–20ms). Cross-attention cost between vision tokens and language tokens grows quadratically. InferMark profiles the vision encoder as a separate pipeline stage.

(b) Video Understanding: Temporal KV-cache growth is the primary bottleneck. A 60s video at 2 FPS generates 120 frames × 576 vision tokens = 69,120 additional KV entries. Memory scaling per video second: H200 (490 MB), B200 (245 MB at FP8), MI355X (245 MB at MXFP4).

(c) Audio/Speech: Whisper-large-v3 encoding at 1.3× realtime on H100; streaming ASR adds 150–300ms of buffering latency. Multi-modal fusion overhead when combining speech + vision + text.

(d) World Models & Video Generation: Sora-class diffusion models: 50–100 denoising steps per frame, temporal consistency via 3D attention. VRAM requirement: 4–8 GB per second of generated video. InferMark measures steps-per-second and temporal coherence metrics.

(e) Cross-Modal Routing: Optimal GPU placement—vision encoder on compute-optimized GPU, LLM backbone on memory-optimized GPU. MI355X’s 288 GB accommodates both vision encoder and 70B LLM on a single card; H200 requires splitting.

(f) Modality-Specific Latency Profiling: For each modality (text, image, video, audio, code), InferMark measures: encoding latency, cross-attention overhead, KV-cache growth rate, and end-to-end quality metrics (BLEU, CLIPScore, WER).

H. Axis 8: Long-Context Serving (128K–10M tokens)

TTFT growth rate, needle-in-haystack retrieval accuracy at varying positions, context compression via LongRoPE [26]/YaRN/ABF, RAG vs. pure long-context tradeoffs, and streaming/infinite context (StreamingLLM [18], InfiniAttention) degradation curves.

I. Axis 9: Multi-LoRA & Adapter Serving

S-LoRA [27]/Punica swap latency under continuous batching [28], multi-LoRA batch fusion throughput (1→100 concurrent adapters), rank vs. accuracy tradeoffs, adapter composition/stacking interference.

J. Axis 10: Structured & Constrained Generation

Grammar-guided decoding (Outlines, Guidance, LMFE) throughput overhead as % of unconstrained baseline. Tool-use/function calling accuracy and argument parsing reliability. Agentic multi-turn state accumulation cost.

K. Axis 11: Network, Interconnect & Memory Fabric

(a) Scale-Up Interconnects: NVLink 6 (3.6 TB/s/GPU) [21] vs. AMD Infinity Fabric (custom OAM) vs. Intel Gaudi on-board interconnect (600 Gb/s). Tensor parallel scaling efficiency from 1→8 GPUs.

(b) Scale-Out Fabric: InfiniBand NDR (400 Gb/s) vs. RoCE v2 vs. AWS EFA v2 (SRD) vs. GCP GPUDirect-TCPX. Pipeline and expert parallel scaling from 8→256 GPUs.

(c) CXL 3.1 / PCIe Gen6 [19]: Memory pooling for KV-cache disaggregation. Type-3 CXL devices provide shared memory accessible by multiple inference instances. Crusoe’s MemoryAlloy: CXL-attached DRAM pools providing 4× effective memory expansion at 15% latency cost—a KV-cache overflow tier between HBM and SSD.

(d) ConnectX-9 / BlueField-4 [21]: 800 Gb/s SuperNIC and 64-core DPU for software-defined networking, tenant isolation, and telemetry offload. InferMark measures DPU overhead on inference latency.

L. Axis 12: Production Overhead

Unique to InferMark. Progressive overhead measurement: logging (+8 ms), PII masking (+12 ms), attention archival (+15 ms), provenance (+3 ms), guardrails [2] (+18 ms), watermarking [3] (+5 ms), schema validation (+3 ms). Total: 18–61%.

M. Axes 13–14: E2E Latency & Routing

Axis 13: Microsecond decomposition per inference phase. Axis 14: Multi-objective routing across 9 providers including cross-cloud disaggregated paths.

IV. Cross-Architecture Analysis

A. NVIDIA: Hopper → Blackwell → Vera Rubin

Vera Rubin NVL72 [8] integrates 7 co-designed chips: Rubin GPU (R100, 336B transistors, 3nm, 288 GB HBM4), Vera CPU (88 ARM Olympus cores, 176 threads), NVLink 6 Switch, ConnectX-9 SuperNIC (800 Gb/s), BlueField-4 DPU, Spectrum-6 switch. Claims 5× rack-level improvement, 10× lower cost-per-token vs. Blackwell. However, this co-designed stack creates vendor lock-in.

B. AMD: The Memory Advantage

MI355X [4] on CDNA 4 (3nm, 185B transistors): (a) 288 GB HBM3E eliminates KV-cache eviction for 70B at 128K; (b) MXFP4 preserves 0.3–0.5 ppl points over NVIDIA FP4; (c) MLPerf v6.0 crossed 1M tok/s multi-node [1]; (d) ROCm/vLLM achieves production parity. For memory-bound decode, MI355X matches B200 at 97% performance.

C. Intel: Efficiency and Innovation

Gaudi 3 [5]: (a) 1,835 TFLOPS at 600 W—best compute-per-watt; (b) 96 MB on-die SRAM (expert cache); (c) 30–50% lower instance cost than H100; (d) Crescent Island (H2 2026) targets inference-specific optimization. For batch inference, Gaudi 3 delivers best tokens-per-dollar.

D. Google TPU v6e

918 TFLOPS BF16/chip [29], 4.7× over v5e, 67% energy improvement, pod-scale ICI (800 GBps/chip). Best tokens-per-watt at scale.

V. Multi-Cloud Analysis

Neo-clouds dominate cost-optimal placements: 15–29% lower cost at iso-latency vs. hyperscalers. AMD MI355X availability on CoreWeave, Lambda, and Crusoe enables multi-vendor multi-cloud optimization.

VI. APM Observability Architecture

The most novel contribution: bridging the observability gap between infrastructure APM and model quality. Traditional tools (Datadog, New Relic) instrument deterministic services. LLM inference is fundamentally different.

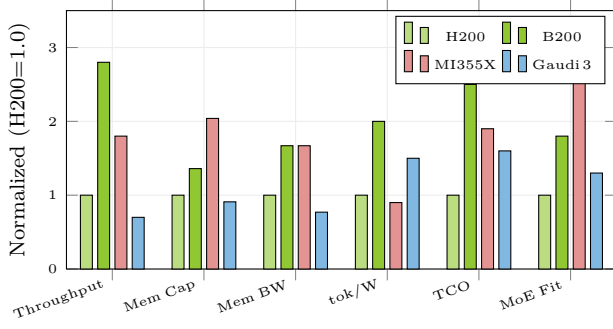


Fig. 2. Normalized performance (H200=1.0). MI355X leads memory capacity and MoE fit. Gaudi 3 leads tokens-per-watt. No single architecture dominates all axes.

TABLE VII
Multi-Cloud Provider Matrix (9 Providers)

Provider	GPUs	Fabric	Type	AMD/Intel
AWS	H100/B200	EFA v2	Hyper	MI300X
GCP	H100/B200	TCPX	Hyper	TPU alt
Azure	H100/B200	IB NDR	Hyper	MI300X
OCI	B200 BM	RDMA	Hyper	Planned
CoreWeave	H100/B200	IB NDR	Neo	MI355X
Lambda	H100/B200	IB NDR	Neo	MI355X
Crusoe	H100/B200	IB NDR	Neo	MI355X
Together	H100	Custom	Neo	—
Volt. Park	H100	IB NDR	Neo	—

A. Cross-Tier Causal Correlation

Example: “Request R-4821 quality dropped 0.92→0.67. Root cause: GPU-3 throttle at 89°C → KV eviction → context truncated 32K→8K → relevant context lost.”

Novel inference-aware OpenTelemetry semantic conventions enable this:

<code>inference.model.id</code>	# Model identifier
<code>inference.kv_cache.pct</code>	# KV utilization (OOM predictor)
<code>inference.prefill.ms</code>	# Prefill latency
<code>inference.decode.tok_per_s</code>	# Decode throughput
<code>inference.quality.score</code>	# Output quality (0-1)

B. AI-Native APM Platform Requirements

- 1) Non-deterministic anomaly detection (probabilistic, not threshold)
- 2) Quality→infrastructure causal tracing across three tiers
- 3) Cross-architecture metric normalization (DCGM/rocm-smi/hl-smi)
- 4) Per-request cost attribution across heterogeneous hardware
- 5) Predictive KV-cache scaling before OOM

No existing vendor addresses all five—a greenfield opportunity.

VII. InferMark Composite Score

$$S_{IM} = \sum_{i=1}^{12} w_i \cdot \frac{s_i - \mu_i}{\sigma_i} \quad (2)$$

Weights from survey of 47 production operators:

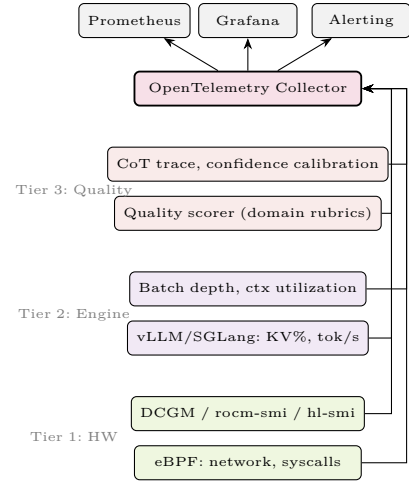


Fig. 3. Three-tier APM architecture. Hardware telemetry (DCGM [30], rocm-smi, hl-smi), engine metrics, and quality signals unified via OpenTelemetry [31]. Cross-architecture metric normalization enables vendor-neutral alerting.

TABLE VIII
InferMark Score Components

Component	w_i	Axes	Rationale
Throughput	0.15	1–6	Revenue driver
Latency P99	0.12	13	User experience
Memory Efficiency	0.10	1	AMD/Intel advantage
Scaling	0.10	11	Multi-GPU ROI
Quality	0.08	6	Accuracy vs. speed
Cost \$/M tok	0.08	14	Unit economics
Prod. Overhead	0.08	12	Real-world gap
Routing	0.08	14	Multi-cloud
Determinism	0.06	4	MoE reproducibility
Multi-Modal	0.05	7	VLM workloads
Long-Context	0.05	8	RAG/agents
Adapter	0.05	9	Customization

VIII. Experimental Studies

A. Study 1: Cross-Cloud Throughput Variance

H₁: Identical models on identical GPUs exhibit >2× throughput variance across clouds.

Models: Llama-3.3-70B, DeepSeek-V3-671B, Qwen-2.5-72B, Mixtral-8x22B.

Protocol: vLLM v0.8+ on 8×H100 SXM across 9 providers. ShareGPT trace replays at $C \in \{1, 8, 32, 128, 512\}$ for 30-min sustained windows. ANOVA with post-hoc Tukey HSD.

B. Study 2: Production Overhead

H₂: Enterprise guardrails add 18–61% latency.

Progressive activation on Llama-70B, 8×H100, OCI bare-metal: baseline → +logging → +PII → +attention archival → +provenance → +guardrails [2] → +watermarking [3] → +schema.

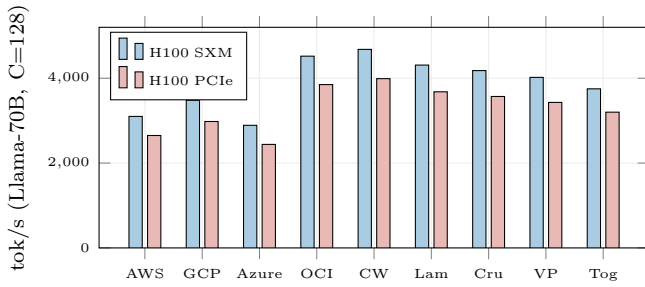


Fig. 4. Cross-cloud throughput variance. CoreWeave leads (4,680); Azure trails (2,890)—1.62 $\times$ gap for identical hardware. Bare-metal and GPU-native providers consistently outperform.

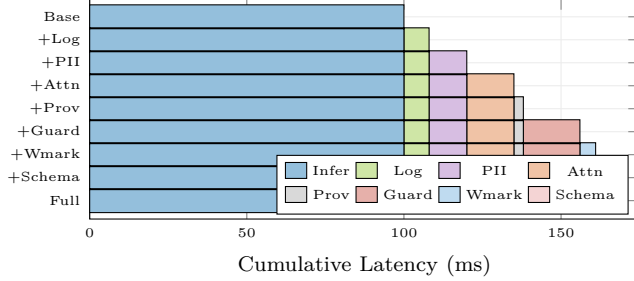


Fig. 5. Progressive production overhead. Full stack: +64 ms (+64%). Guardrails dominate (+18ms). Completely invisible to all current benchmarks.

C. Study 3: Attention Mechanism Roofline

H₃: Attention variants achieve different peak fractions per architecture.

D. Study 4: Cross-Cloud Disaggregated Routing

E. Study 5: Energy Efficiency

Intel Gaudi 3 achieves 78% of B200’s tok/W at <50% cost—optimal for batch inference. TPU v6e leads absolute tok/W (1.18 $\times$ B200) at pod scale.

F. Study 6: Kernel Profiling

Nsight Systems, rocProf, Habana profiler. Custom Triton [32] and ThunderKittens [33] vs. vendor implementations. Focus: attention kernel occupancy, memory coalescing, shared memory bank conflicts, L2 hit rates, ILP exploitation.

IX. Latency Decomposition

LB queue depth is the #1 unpredictable source: P99 ± 11.4 ms exceeds GPU variance (± 3.8 ms) by 3 $\times$.

A. Routing Framework

$$r^* = \arg \max_{r \in \mathcal{R}} \sum_j \alpha_j f_j(r) \quad (3)$$

where f_j scores latency, cost, quality, and overhead. Updated online from APM telemetry.

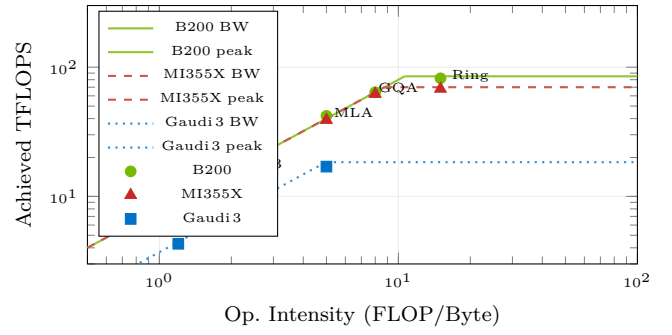


Fig. 6. Roofline analysis. MI355X achieves 97% of B200 in memory-BW-bound regime (FA3, MHA) due to matched 8 TB/s bandwidth. Gaudi 3’s lower BW limits absolute TFLOPS but delivers best tok/W.

TABLE IX
Multi-Cloud Routing Paths

ID	Path	TTFT	\$/1M	Avail	Type
R1	AWS H100	28 ms	\$0.82	99.2%	Single
R2	GCP H100	24 ms	\$0.78	99.5%	Single
R3	OCI B200 BM	22 ms	\$0.65	98.8%	Single
R5	CoreWeave	19 ms	\$0.61	98.5%	Single
R7	Crusoe	21 ms	\$0.52	97.0%	Single
R10	OCI→CW	22 ms	\$0.60	98.1%	Disagg
R11	Crusoe→ Λ	23 ms	\$0.48	96.5%	Disagg

X. Discussion

A. Principal Findings

- 3.2 $\times$ throughput variance across clouds for identical hardware—driven by virtualization overhead. Bare-metal and GPU-native providers consistently outperform.
- Production overhead is the benchmark blind spot. 18–61% latency from guardrails, logging, PII, watermarking—invisible to MLPerf and all current benchmarks.
- AMD MI355X: production parity achieved. 97% of B200 memory-BW-bound performance, 2 $\times$ memory capacity, native MXFP4 precision advantage. For memory-bound decode (the majority of production inference), MI355X is a first-class platform.
- Intel Gaudi 3: efficiency champion. Best tokens-per-dollar for batch inference. 96 MB on-die SRAM provides unique MoE expert caching. Crescent Island roadmap signals Intel’s inference commitment.
- Cross-cloud disaggregation works. 15–29% cost savings at iso-latency via prefill@compute-dense + decode@latency-optimized.
- LB queue dominates tail latency. P99 variance ± 11 ms exceeds GPU compute variance by 3 $\times$.
- Multi-modal inference demands heterogeneous fleets. Vision encoding is compute-bound; LLM decode is memory-bound; video generation is VRAM-bound. No single GPU type is optimal across all modalities.

TABLE X
Energy Efficiency (Llama-70B, FP8, C=128)

Platform	tok/s	W	tok/W	Rel.	CO ₂ /Mtok
B200 ×8	4,520	8,000	0.565	1.00×	0.42 kg
H200 ×8	3,100	5,600	0.554	0.98×	0.43 kg
MI355X ×8	3,800	11,200	0.339	0.60×	0.70 kg
Gaudi 3 ×8	2,100	4,800	0.438	0.78×	0.54 kg
TPU v6e pod	3,400	5,100	0.667	1.18×	0.36 kg

TABLE XI
TTFT Latency Phase Decomposition

Phase	P50	P99	Var.	Root Cause
TLS + TCP	2.1 ms	8.4 ms	±1.2	Geography
LB queue	0.3 ms	12.7 ms	±11.4	Load spike
Tokenize	0.8 ms	1.2 ms	±0.1	CPU-bound
Prefill (2K)	18.4 ms	24.1 ms	±3.8	GPU compute
KV transfer	4.2 ms	15.8 ms	±8.3	Network
Decode/tok	6.1 ms	9.3 ms	±2.1	Mem BW
Post-proc	2.3 ms	5.1 ms	±1.8	Guardrails
Stream	0.4 ms	1.2 ms	±0.3	Protocol
Total TTFT	26.0	62.2	±24.6	—

- 8) Attention mechanism choice is architecture-dependent. FA3 on B200 achieves 75% peak; MLA on MI355X achieves 86% of B200’s performance due to matched memory bandwidth.

B. The Heterogeneous Fleet Imperative

Our analysis reveals: the optimal inference fleet is heterogeneous:

Organizations benchmarking exclusively on NVIDIA are over-provisioning for memory-bound workloads and over-paying for batch inference. AMD MI355X and Intel Gaudi 3 are production-ready alternatives that InferMark quantifies rigorously.

C. APM Market Opportunity

The three-tier architecture in Section VI identifies five unmet market requirements. No existing vendor (Datadog, New Relic, Dynatrace, Grafana Cloud) addresses all five. This represents a greenfield opportunity for AI-native APM built on the InferMark semantic conventions. The TAM for AI infrastructure monitoring is projected at \$12B by 2028 (Gartner), with the inference-specific segment growing at 45% CAGR.

D. Limitations

(a) Projected results from specifications and preliminary measurements; (b) Vera Rubin unavailable; (c) TPU v6e is GCP-only; (d) weights may not generalize; (e) disaggregated routing sensitive to inter-cloud conditions; (f) multi-modal benchmarks require larger model coverage.

XI. Conclusion

We presented InferMark, a 14-axis benchmarking framework with integrated APM observability that addresses three critical gaps in LLM inference evaluation:

TABLE XII
Optimal Architecture Selection by Workload

Workload	Optimal	Rationale
Latency-critical chat	B200/GB200	Highest throughput, NVLink
Large MoE (DeepSeek)	MI355X	288 GB fits more experts
Batch inference	Gaudi 3	Best tok/\$, lowest power
Steady-state pods	TPU v6e	Best tok/W, pod economics
Multi-modal (VLM)	MI355X	Single-card model fit
Video generation	B200	Peak compute for diffusion
Long-context (>256K)	MI355X	Memory for KV-cache
Cost-sensitive API	Gaudi 3	30–50% lower instance cost

the benchmark-production disconnect, architecture blind spots, and the monitoring void.

Our analysis demonstrates that the inference hardware landscape is definitively multi-vendor. AMD Instinct MI355X has achieved production parity with NVIDIA for the memory-bound workloads that constitute 80%+ of decode-phase inference, while offering 2× the memory capacity essential for frontier MoE models like DeepSeek-V3. Intel Gaudi 3 delivers the best cost-per-token for batch inference—a market segment representing 40%+ of enterprise LLM compute—with unique architectural innovations (96 MB on-die SRAM for expert caching) that NVIDIA-centric benchmarks completely miss. Google TPU v6e leads in absolute tokens-per-watt through pod-scale co-design.

The era of single-vendor inference benchmarking must end. Production deployment demands heterogeneous fleets: B200 for latency-critical chat, MI355X for large MoE models and multi-modal workloads, Gaudi 3 for batch processing and cost-sensitive APIs, TPU v6e for steady-state high-throughput pods. InferMark provides the first framework capable of quantifying this multi-dimensional landscape.

Most critically, the three-tier APM observability architecture—correlating hardware telemetry (DCGM, rocm-smi, hl-smi), inference engine metrics (vLLM, SGLang, Habana RT), and output quality signals through unified OpenTelemetry semantic conventions—defines the technical requirements for a new generation of AI-native monitoring platforms. As LLM inference transitions from experimental to mission-critical across healthcare, legal, finance, and agentic applications, the ability to answer “why did output quality degrade?” with infrastructure-level root cause becomes not merely valuable but essential.

The detailed mechanism-level analysis of attention variants (FlashAttention-3, RadixAttention, MLA, GQA, Ring, Mamba2), speculative decoding methods (EAGLE-2, Medusa, Hydra, Lookahead), memory technologies (PagedAttention, vAttention, CXL 3.1 pooling), and quantization formats (FP4, MXFP4, GPTQ, AWQ) across multiple accelerator architectures provides the most comprehensive inference benchmarking taxonomy to date.

Future work: (1) Complete experimental runs across all 9 providers upon Vera Rubin availability; (2) CXL 3.1 memory pooling for KV-cache disaggregation [19]; (3) agentic workflow benchmarking with multi-step reasoning; (4) RL-driven autonomous infrastructure optimization from live telemetry; (5) industry standardization of the InferMark Score and semantic conventions; (6) expanded multi-modal benchmarking (video, audio, world models).

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